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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

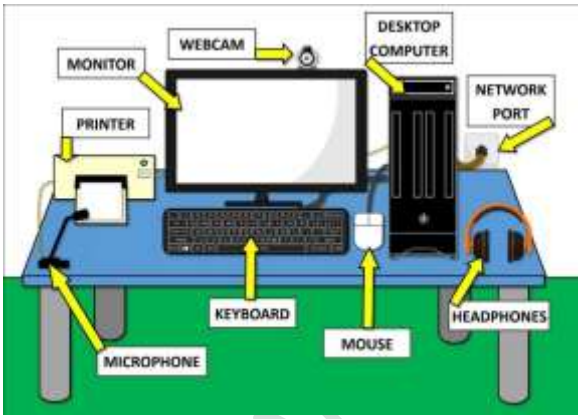
Level 2 – Young Technologist (7th–8th Standard)

Class 2: Computer Basics



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Learning Objectives

By the end of this lesson, students will be able to:

- Understand what a computer is.
- Identify the main parts of a computer.
- Explain how a computer works.
- Understand the characteristics and uses of computers.
- Recognize different types of computers.
- Perform basic computer operations safely.

What is a Computer?

A **computer** is an **electronic device** that accepts data, processes it according to instructions, stores it, and produces useful information.

Simple Definition

A computer is a smart machine that helps us learn, solve problems, communicate, create documents, play educational games, and perform many daily tasks quickly and accurately.

Where Do We Use Computers?

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Computers are used in many places:

Place	Use of Computer
🏫 School	Online classes, exams, projects
🏥 Hospital	Patient records, medical reports
🏦 Bank	Money transfers and account management
🛒 Shopping Mall	Billing and inventory
✈️ Airport	Ticket booking and flight information
🏠 Home	Learning, entertainment, communication
🏢 Office	Documents, presentations, emails
🚓 Police Station	Maintaining records and surveillance

Characteristics of a Computer

1. Speed

A computer performs millions of calculations in a very short time.

Example: It can calculate the total marks of 1,000 students within seconds.

2. Accuracy

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A computer gives correct results when the instructions and data are correct.

3. Storage

A computer can store a large amount of information such as photos, videos, documents, and games.

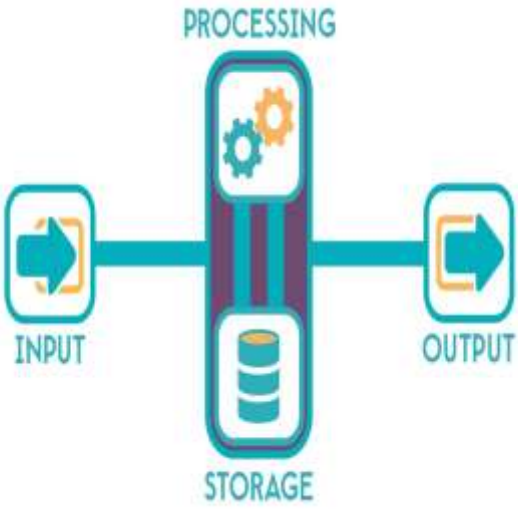
4. Diligence

A computer never gets tired and can work continuously.

5. Versatility

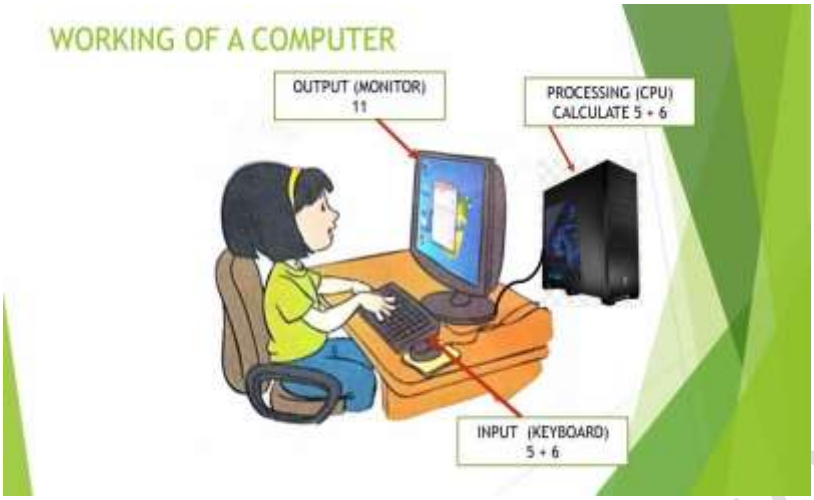
A single computer can be used for studying, drawing, coding, gaming, video editing, and many other tasks.

How Does a Computer Work





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A computer follows the **IPO Cycle**:

1. **Input** – Data is entered using devices like a keyboard or mouse.
2. **Processing** – The CPU processes the data.
3. **Output** – The result is displayed on a monitor or printed.
4. **Storage** – Information is saved for future use.

Example

- Input: Type **5 + 3**
- Processing: CPU performs the calculation.
- Output: **8**
- Storage: Save the result in a file.

Main Parts of a Computer

Part	Function
Monitor	Displays text, images, and videos
Keyboard	Used for typing
Mouse	Used to point, click, and select
CPU (Central Processing Unit)	Processes instructions; called the "brain" of the computer

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Part	Function
🔊 Speakers	Produce sound
🎤 Microphone	Records voice
📷 Webcam	Captures photos and videos
🖨️ Printer	Prints documents on paper

What is the CPU?

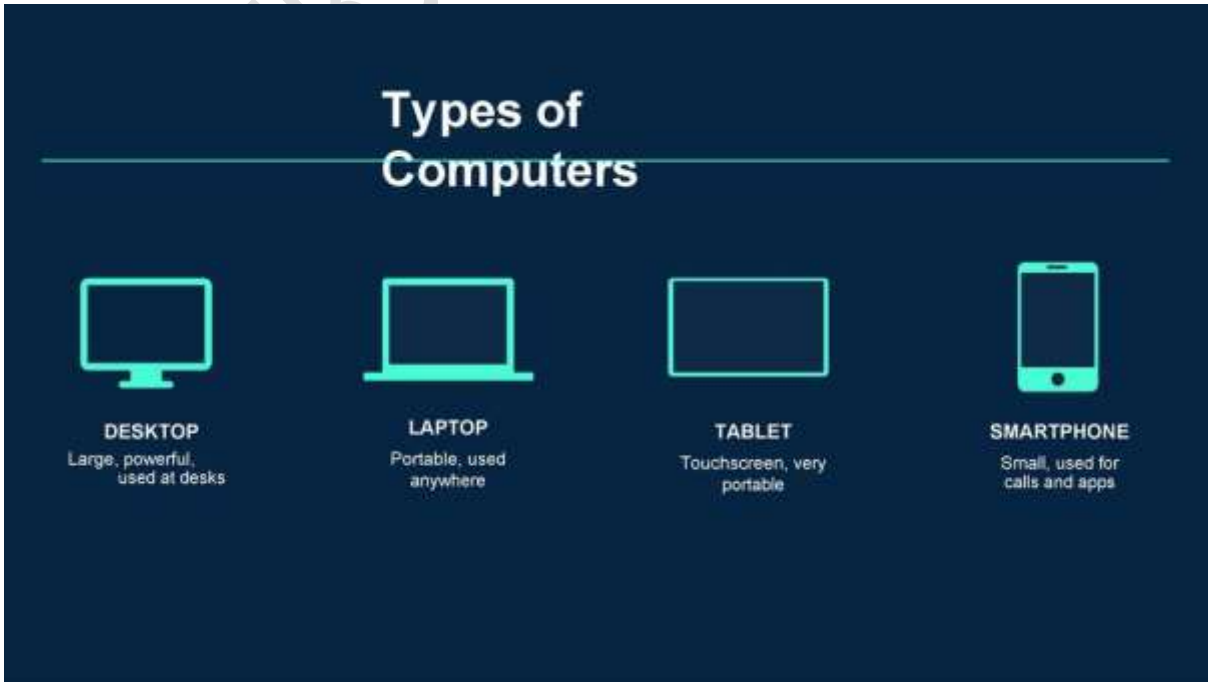
The **Central Processing Unit (CPU)** is the **brain of the computer**.

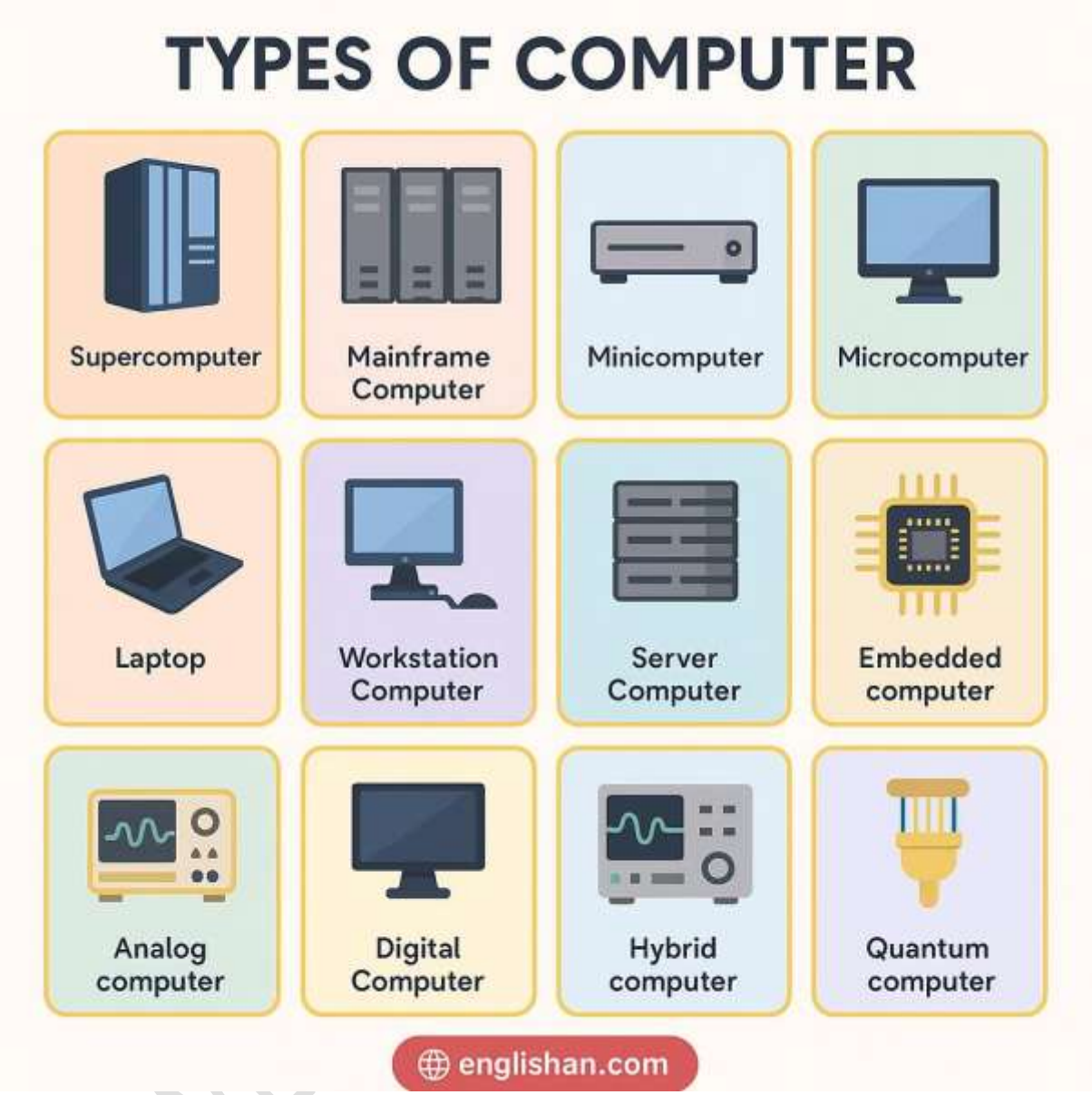
It:

- Performs calculations.
- Executes instructions.
- Controls all computer operations.
- Coordinates with other hardware.

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Types of Computers





1. Desktop Computer

- Used in homes, schools, and offices.
- Powerful and easy to upgrade.

2. Laptop

- Portable and battery-powered.
- Easy to carry.

3. Tablet

- Touchscreen device.
- Lightweight and portable.

4. Smartphone

- Small computer with calling and internet features.

5. Supercomputer

- Extremely powerful.
- Used for weather forecasting, scientific research, and space exploration.

Hardware and Software

Hardware	**Software**
Physical parts you can touch	Programs and applications you use
Keyboard	MS Word
Mouse	Paint
Monitor	Python
Printer	Web Browser

Basic Computer Operations

Students should practice:

- Switching on the computer.
- Logging in (if required).
- Opening an application (e.g., Paint or Calculator).
- Typing using the keyboard.
- Using the mouse to click and drag.
- Closing applications properly.
- Shutting down the computer safely.

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Advantages of Computers

- Fast calculations
- Accurate results
- Easy communication
- Online learning
- Entertainment
- Creativity
- Information storage
- Supports scientific research

Safety Rules in the Computer Lab

- Sit with good posture.
- Keep hands clean.
- Do not eat or drink near computers.
- Handle the keyboard and mouse gently.
- Ask the teacher before installing software.
- Save your work regularly.
- Shut down the computer properly.

Practical Activities

Practical Activity 1 – Identify the Parts

Objective: Learn the main computer components.

Instructions:

1. Observe a desktop or laptop.
2. Point to and name:
    - Monitor
    - Keyboard

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- Mouse
- CPU/System Unit
- Speakers
- Webcam (if available)

3. Explain one function of each part.

Practical Activity 2 – IPO Cycle

Fill in the table:

Activity	Input	Processing	Output
Typing your name	Keyboard	CPU processes letters	Name appears on the monitor
Playing a song	Mouse	CPU opens music file	Sound from speakers
Drawing in Paint	Mouse	CPU processes drawing	Picture appears on screen

Practical Activity 3 – Hardware or Software?

Write **H** for Hardware and **S** for Software.

Item	Answer
Monitor	H
Keyboard	H
Python	S
MS Paint	S
Printer	H
Web Browser	S
Mouse	H
Calculator App	S

Classroom Activity

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Match the Device to Its Function

Device	Function
Keyboard	Typing text
Mouse	Pointing and clicking
Monitor	Displaying information
Printer	Printing documents
Speakers	Playing sound

Quick Quiz

1. What is a computer?

A) A toy

B) An electronic device that processes data ✅

C) A book

D) A table

2. Which part is called the brain of the computer?

A) Mouse

B) Keyboard

C) CPU ✅

D) Monitor

3. Which device is used for typing?

A) Keyboard ✅

B) Mouse

C) Printer

D) Speaker

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4. Which of the following is software?

A) Monitor

B) Mouse

C) Python ✅

D) Keyboard

5. Which stage of the IPO cycle displays the result?

A) Input

B) Output ✅

C) Storage

D) Process

Home Assignment

1. Draw and label the main parts of a computer.
2. Write **five uses of computers** in everyday life.
3. List **three hardware** devices and **three software** applications.
4. Observe a computer at home or school and identify its parts.

Lesson Summary

In this lesson, students learned:

- The meaning of a computer.
- Characteristics of computers.
- The IPO (Input–Process–Output) cycle.
- Main parts of a computer and their functions.
- Types of computers.
- Difference between hardware and software.
- Safe computer lab practices.

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- Basic hands-on computer operations.

Learning Outcomes

By the end of this class, students can:

- Define a computer.
- Identify the main parts of a computer.
- Explain the IPO cycle.
- Distinguish between hardware and software.
- Describe different types of computers.
- Use a computer safely and responsibly.
- Perform basic computer operations with confidence.

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